

Internship report

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1 Introduction

I did, between June 06 2025, and July 11 2025, an internship under the supervision of Mr. Jun Sekine. The internship took place at Osaka University and the topic was *Optimal stochastic control for Brownian models*.

The internship was centred around an article : *Trading to Stops* by L.C.G Rogers and Nora Imkeller. The aim of the internship was to understand and then generalize the results of this article. The internship went as follows : first I studied all the notions needed to apprehend the article, then with all this knowledge I was able to read the article and finally, I tried to find new results. This report will follow a similar approach even though due to a size constraint it will be difficult to be exhaustive.

2 Brownian Motion

In this section, we define Brownian motion and prove its existence. We will also demonstrate some of its properties.

2.1 Definition and existence

Definition 2.1 (Brownian Motion). *A Brownian motion is defined as the real-valued stochastic process*

$$B = (\Omega, (\mathcal{F}_t)_{t \geq 0}, (B_t)_{t \geq 0}, \mathbb{P})$$

such that:

- i) $B_0 = 0$ $\mathbb{P}$ -a.s. (almost surely)
- ii) $\forall 0 \leq s \leq t$, the random variable $B_t - B_s$ is independent of $\mathcal{F}_s$
- iii) $\forall 0 \leq s \leq t$, $B_t - B_s$ follows the distribution $\mathcal{N}(0, t - s)$

Proposition 2.1. *A Brownian motion is a process with independent increments.*

$\forall t_0 \leq \dots \leq t_n$, the random variables $(B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}})$ are independent and also independent of $\mathcal{F}_{t_0}$.

Proof. Let $t_0 \leq \dots \leq t_n$.

Let $A_1, \dots, A_n$ be Borel sets in $\mathbb{R}$, and let $A \in \mathcal{F}_{t_0}$.

Let B be a Brownian motion.

From 2.1.ii), we have:

$$\forall i \in \{1, \dots, n\}, (B_{t_i} - B_{t_{i-1}}) \text{ is independent of } \mathcal{F}_{t_{i-1}}.$$

But $\mathcal{F}_{t_{i-1}} = \sigma(B_r; r \leq t_{i-1})$.

Hence:

$$\forall i \in \{1, \dots, n\}, (B_{t_i} - B_{t_{i-1}}) \text{ is independent of } (B_{t_k} - B_{t_{k-1}})_{k \leq i} \text{ and of } \mathcal{F}_{t_0}.$$

Then, by backward induction on $i \in \{1, \dots, n\}$, we have:

$$\begin{aligned} & \mathbb{P}(B_{t_n} - B_{t_{n-1}} \in \Gamma_n, \dots, B_{t_1} - B_{t_0} \in \Gamma_1, A) \\ &= \mathbb{P}(B_{t_n} - B_{t_{n-1}} \in \Gamma_n) \mathbb{P}(B_{t_{n-1}} - B_{t_{n-2}} \in \Gamma_{n-1}, \dots, B_{t_1} - B_{t_0} \in \Gamma_1, A) \\ &= \left(\prod_{i=1}^n \mathbb{P}(B_{t_i} - B_{t_{i-1}} \in \Gamma_i) \right) \mathbb{P}(A) \end{aligned}$$

We have thus shown that $(B_{t_i} - B_{t_{i-1}})_{1 \leq i \leq n}$ are independent of $\mathcal{F}_{t_0}$.

By taking $A = \Omega$, we have also shown the mutual independence of $(B_{t_i} - B_{t_{i-1}})_{1 \leq i \leq n}$.

This completes the proof. □

Theorem 2.1 (Characterization of Brownian Motion). *i) A Brownian motion B satisfies the following properties:*

a) $B_0 = 0$ $\mathbb{P}$ -a.s.

b) $\forall 0 \leq t_1 \leq \dots \leq t_n$, the vector $(B_{t_1}, \dots, B_{t_n})$ is a centered Gaussian vector.

c) $\forall s, t \geq 0$, $\mathbb{E}(B_s B_t) = \min(s, t)$

That is, B is a centered real-valued Gaussian process, and its covariance function is given by $\Gamma(s, t) = \min(s, t)$.

ii) Conversely, if B is a process that satisfies properties a), b), and c), then, denoting by $(\mathcal{F}_t)_{t \geq 0}$ its natural filtration, we have:

$$B = (\Omega, (\mathcal{F}_t)_{t \geq 0}, (B_t)_{t \geq 0}, \mathbb{P})_t$$

is a Brownian motion. Moreover, it is natural.

Proof of Point i). Let us first prove part i)

Let $B = (\Omega, (\mathcal{F}_t)_{t \geq 0}, (B_t)_{t \geq 0}, \mathbb{P})$ be a Brownian motion.

From the definition of Brownian motion (2.1), property a) holds directly.

It remains to prove b) and then c).

We will prove, by induction on $n \in \mathbb{N}$, the following property:

$$\forall a \in \mathbb{R}^n, \forall 0 \leq t_1 \leq \dots \leq t_n, a_1 B_{t_1} + \dots + a_n B_{t_n} \text{ is a normally distributed random variable. } (H_n)$$

Base case: $n = 1$.

Let $a_1 \in \mathbb{R}$ and $t_1 \in \mathbb{R}_+$.

From the definition (2.1.i)), we know that

$$B_0 = 0 \quad \mathbb{P}\text{-a.s.}$$

Therefore:

$$a_1 B_{t_1} = a_1(B_{t_1} - B_0)$$

And by (2.1.iii)), we have:

$$a_1 B_{t_1} \sim \mathcal{N}(0, a_1^2(t_1 - 0))$$

Thus, $a_1 B_{t_1}$ is a normally distributed random variable.

Inductive step: Let $n \in \mathbb{N}$. We now show that $H_n \implies H_{n+1}$.

Let $a \in \mathbb{R}^{n+1}$ and $0 \leq t_1 \leq \dots \leq t_{n+1}$ be real numbers.

From (2.1.ii)), we know that the random variables

$$a_1 B_{t_1} + \dots + (a_n + a_{n+1}) B_{t_n} \quad \text{and} \quad a_{n+1}(B_{t_{n+1}} - B_{t_n})$$

are independent.

Moreover, by (H_n) , both of them are normally distributed.

Since the sum of two independent Gaussian random variables is Gaussian, we have:

$$a_1 B_{t_1} + \dots + a_{n+1} B_{t_{n+1}} \text{ is a normally distributed random variable.}$$

Thus, $H_n \implies H_{n+1}$.

Conclusion: By induction, B is a Gaussian process. Since all B_t are centered, B is a centered Gaussian process. This proves point b).

It remains to prove point c).

Let $0 \leq s \leq t$.

From 2.1.ii) and 2.1.iii), we have:

$$\mathbb{E}[B_s(B_t - B_s)] = \mathbb{E}[B_s]\mathbb{E}[B_t - B_s] = 0$$

Hence:

$$\begin{aligned} \mathbb{E}[B_s B_t] &= \mathbb{E}[B_s(B_t - B_s) + B_s^2] \\ &= 0 + \mathbb{E}[B_s^2] \\ &= \mathbb{E}[(B_s - B_0)^2] - 0 \\ &= \mathbb{E}[(B_s - B_0)^2] - \mathbb{E}[B_s - B_0]^2 \\ &= \text{Var}[B_s - B_0] \\ &= s \\ &= \min(s, t) \end{aligned}$$

The last equalities follow from the fact that $(B_s - B_0)$ is distributed as $\mathcal{N}(0, s - 0)$ according to 2.1.iii).

This concludes the proof of point $c)$ and thus of part $i)$. □

Proof of Point ii). Let us now prove part $ii)$.

Let B be a process satisfying properties $a)$, $b)$, and $c)$ of 2.1.i).

Property $a)$ is the same as 2.1.i).

Let $0 \leq s \leq t$.

From $b)$, we know that (B_s, B_t) is a centered Gaussian vector.

Therefore, B_s and B_t both follow a centered normal distribution.

Consequently, the linear combination $B_t - B_s$ is normally distributed.

Moreover, $B_t - B_s$ is centered because B_t and B_s are centered and independent.

Let us now compute the variance of $B_t - B_s$.

We have:

$$\begin{aligned} \text{Var}[B_t - B_s] &= \mathbb{E}[(B_t - B_s - \mathbb{E}[B_t - B_s])^2] \\ &= \mathbb{E}[(B_t - B_s - 0)^2] \\ &= \mathbb{E}[B_t^2] + \mathbb{E}[B_s^2] - 2\mathbb{E}[B_s B_t] \end{aligned}$$

And from $c)$, we know that:

$$\begin{aligned} \mathbb{E}[B_t^2] + \mathbb{E}[B_s^2] - 2\mathbb{E}[B_s B_t] &= t + s - 2\min(t, s) \\ &= t + s - 2s \\ &= t - s \end{aligned}$$

Thus, $\text{Var}[B_t - B_s] = t - s$.

Therefore, $B_t - B_s \sim \mathcal{N}(0, t - s)$.

We have established 2.1.iii).

Now, let $0 \leq r \leq s \leq t$.

Let us compute the covariance of $B_t - B_s$ and B_r .

We have:

$$\begin{aligned} \text{Cov}[B_t - B_s, B_r] &= \mathbb{E}[(B_t - B_s - \mathbb{E}[B_t - B_s])(B_r - \mathbb{E}[B_r])] \\ &= \mathbb{E}[(B_t - B_s - 0)(B_r - 0)] \\ &= \mathbb{E}[B_t B_r] - \mathbb{E}[B_s B_r] \end{aligned}$$

And from $c)$, we know that:

$$\begin{aligned} \mathbb{E}[B_t B_r] - \mathbb{E}[B_s B_r] &= \min(t, r) - \min(s, r) \\ &= r - r \\ &= 0 \end{aligned}$$

Therefore, by the theorem on the equivalence between zero covariance and independence for components of a Gaussian vector, $B_t - B_s$ and B_r are independent.

Thus, $(B_t - B_s, B_r)$ is a Gaussian vector with a diagonal covariance matrix.

Consequently, $B_t - B_s$ and B_r are independent for any $r \leq s$.

It follows that $B_t - B_s$ is independent of the σ -algebra $\mathcal{F}_s = \sigma(B_r; r \leq s)$.

Hence, we have established 2.1.ii).

Therefore, B is a Brownian motion with respect to its natural filtration, which proves part ii). $\square$

Theorem 2.2 (Existence of Brownian Motion). *There exists a probability measure $\mathbb{P}$ defined on $\mathbb{R}^+$ endowed with its Borel σ -algebra $B_{\mathbb{R}}^{\otimes \mathbb{R}^+}$ such that the process $(X_t)_{t \geq 0}$ of its coordinate mappings is a natural Brownian motion.*

Proof. We will use theorem ?? to construct a real Gaussian process that satisfies the characterization of Brownian motion given in 2.1.

To do this, we will prove the following result by induction on $n \in \mathbb{N}^*$:

$$\forall 0 \leq t_1 \leq \dots \leq t_n, \Gamma = (\min(t_i, t_j))_{1 \leq i, j \leq n} \text{ is positive semidefinite.} \quad (2.1)$$

We will prove that Γ is positive semidefinite by showing, via induction on $n \in \mathbb{N}^*$, that for any Γ satisfying the conditions of 2.1, we have:

$$\forall v = (v_1, \dots, v_n) \in \mathbb{R}^n, \quad \langle v | \Gamma_n v \rangle \geq t_1(v_1^2 + \dots + v_n^2) \geq 0 \quad (H_n)$$

Base case: $n = 1$.

The result is obvious when $n = 1$.

Inductive step

Assume H_n holds for some $n \in \mathbb{N}^*$.

Let $0 \leq t_1 \leq \dots \leq t_{n+1}$ and let Γ be the corresponding matrix.

Writing Γ explicitly, we have:

$$\Gamma = \begin{pmatrix} t_1 & t_1 & t_1 & t_1 \\ t_1 & t_2 & \dots & t_2 \\ t_1 & \vdots & \ddots & \vdots \\ t_1 & t_2 & \dots & t_{n+1} \end{pmatrix} \quad (2.2)$$

We observe that the first row and the first column of the matrix contain only t_1 .

We also note that the submatrix

$$\Gamma_n = (\min(t_i, t_j))_{2 \leq i, j \leq n+1}$$

satisfies 2.1 for the values $t_2 \leq \dots \leq t_{n+1}$.

Let $u = (u_1, \dots, u_{n+1}) \in \mathbb{R}^{n+1}$.

Set $v = (u_2, \dots, u_{n+1})$.

$$\begin{aligned}\langle u|\Gamma u\rangle &= t_1 u_1^2 + 2t_1 u_1 u_2 + 2t_1 u_1 u_3 + \cdots + 2t_1 u_1 u_{n+1} + \langle v|\Gamma_n v\rangle \\ &= t_1 u_1^2 + 2t_1 u_1(u_2 + \cdots + u_{n+1}) + \langle v|\Gamma_n v\rangle\end{aligned}$$

And from H_n , we have:

$$\langle v|\Gamma_n v\rangle \geq t_2(u_2^2 + \cdots + u_{n+1}^2)$$

Moreover, since $t_2 \geq t_1$:

$$\langle v|\Gamma_n v\rangle \geq t_1(u_2^2 + \cdots + u_{n+1}^2)$$

Thus, returning to the computation of $\langle u|\Gamma u\rangle$:

$$\begin{aligned}\langle u|\Gamma u\rangle &\geq t_1 u_1^2 + 2t_1 u_1(u_2 + \cdots + u_{n+1}) + t_1(u_2^2 + \cdots + u_{n+1}^2) \\ &= t_1(u_1^2 + 2u_1(u_2 + \cdots + u_{n+1}) + u_2^2 + \cdots + u_{n+1}^2) \\ &= t_1(u_1 + u_2 + \cdots + u_{n+1})^2 \\ &\geq 0\end{aligned}$$

Therefore, $H_n \implies H_{n+1}$.

Conclusion

By induction, Γ satisfies 2.1.

Hence, Γ is a symmetric positive semidefinite matrix.

By theorem ??, with $T = \mathbb{R}^+$ and $m : t \mapsto 0$, we obtain a real Gaussian process that satisfies the characterization of Brownian motion given in 2.1.

Therefore, we have proved the existence of Brownian motion. □

2.2 Some properties

Theorem 2.3 (Invariances of Brownian Motion). *Let $B = (\Omega, (\mathcal{F}_t)_{t \geq 0}, (B_t)_{t \geq 0}, \mathbb{P})$ be a Brownian motion.*

Let $s > 0$ and $c > 0$ be fixed real numbers.

Then the following processes are also Brownian motions:

- i) $B_t^{(1)} = -B_t$
- ii) $B_t^{(2)} = B_{t+s} - B_s$
- iii) $B_t^{(3)} = cB_{\frac{t}{c^2}}$
- iv) $B_0^{(4)} = 0$ and, $\forall t > 0$, $B_t^{(4)} = tB_{\frac{1}{t}}$

Proof of Points i), ii) and iii). We will show that $B^{(1)}$, $B^{(2)}$, and $B^{(3)}$ satisfy the definition 2.1 of a Brownian motion.

Proof of i) We have $B_0^{(1)} = -B_0 = -0 = 0$, hence 2.1.i) holds.

Let $0 \leq k \leq t$.

We compute:

$$B_t^{(1)} - B_k^{(1)} = -(B_t - B_k)$$

From 2.1.ii), $B_t - B_k$ is independent of $\mathcal{F}_k$.

By the continuity of the transformation $x \mapsto -x$, we deduce that:

$$-(B_t - B_k) = B_t^{(1)} - B_k^{(1)} \text{ is also independent of } \mathcal{F}_k.$$

Thus, 2.1.ii) holds.

Moreover, we have:

$$B_t^{(1)} - B_k^{(1)} = -(B_t - B_k), \text{ and } B_t - B_k \sim \mathcal{N}(0, t - k).$$

Therefore, $-(B_t - B_k) \sim \mathcal{N}(0, (-1)^2(t - k)) = \mathcal{N}(0, t - k)$.

Hence, 2.1.iii) is satisfied.

Consequently, $B^{(1)}$ is a Brownian motion with respect to the filtration $(\mathcal{F}_t)_{t \geq 0}$.

Proof of ii) We have $B_0^{(2)} = B_{0+s} - B_s = B_s - B_s = 0$, hence 2.1.i) holds.

Let $0 \leq k \leq t$. Then:

$$B_t^{(2)} - B_k^{(2)} = (B_{t+s} - B_s) - (B_{k+s} - B_s) \tag{2.3}$$

$$= B_{t+s} - B_{k+s} \tag{2.4}$$

From 2.1.ii), $B_{t+s} - B_{k+s}$ is independent of $\mathcal{F}_{k+s}$,

and thus also independent of $\mathcal{F}_k$ since $s \geq 0$.

Therefore, $B_t^{(2)} - B_k^{(2)}$ is independent of $\mathcal{F}_k$, which gives 2.1.ii).

Moreover:

$$B_t^{(2)} - B_k^{(2)} = B_{t+s} - B_{k+s}$$

and from 2.1.iii),

$$B_{t+s} - B_{k+s} \sim \mathcal{N}(0, (t+s) - (k+s)) = \mathcal{N}(0, t - k).$$

Hence, 2.1.iii) is satisfied.

Thus, $B^{(2)}$ is a Brownian motion with respect to the filtration $(\mathcal{F}_t)_{t \geq 0}$.

Proof of iii) We have $B_0^{(3)} = cB_{\frac{0}{c^2}} = cB_0 = c \times 0 = 0$, hence 2.1.i) holds.

Let $0 \leq k \leq t$.

We compute:

$$B_t^{(3)} - B_k^{(3)} = c(B_{\frac{t}{c^2}} - B_{\frac{k}{c^2}})$$

From 2.1.ii), $B_{\frac{t}{c^2}} - B_{\frac{k}{c^2}}$ is independent of $\mathcal{F}_{\frac{k}{c^2}}$.

Therefore, $B_t^{(3)} - B_k^{(3)}$ is independent of $\mathcal{F}_{\frac{k}{c^2}}$.

Hence, 2.1.ii) holds for the filtration $(\mathcal{F}_{\frac{t}{c^2}})_{t \geq 0}$.

Moreover:

$$B_t^{(3)} - B_k^{(3)} = c(B_{\frac{t}{c^2}} - B_{\frac{k}{c^2}})$$

and from 2.1.iii), $B_{\frac{t}{c^2}} - B_{\frac{k}{c^2}} \sim \mathcal{N}\left(0, \frac{t-k}{c^2}\right)$.

$$\text{Therefore, } c(B_{\frac{t}{c^2}} - B_{\frac{k}{c^2}}) \sim \mathcal{N}\left(0, \frac{c^2(t-k)}{c^2}\right) = \mathcal{N}(0, t-k).$$

Hence, 2.1.iii) holds.

Consequently, $B^{(3)}$ is a Brownian motion with respect to the filtration $(\mathcal{F}_{\frac{t}{c^2}})_{t \geq 0}$.

□

Proof of Point iv). We will show that $B^{(4)}$ satisfies the characterization given in theorem 2.1.

By definition, $B_0^{(4)} = 0$,
hence 2.1.i).a) is satisfied.

Since B is a Brownian motion, it satisfies 2.1.i).b). Thus, for any $0 \leq t_1 \leq \dots \leq t_n$,

the vector $(B_{\frac{1}{t_n}}, \dots, B_{\frac{1}{t_1}})$ is a centered Gaussian vector.

Therefore, $(t_n B_{\frac{1}{t_n}}, \dots, t_1 B_{\frac{1}{t_1}})$ is also a centered Gaussian vector.

Hence, 2.1.i).b) holds.

Let $a > 0$ and $b > 0$. We compute:

$$\begin{aligned} \mathbb{E}[B_a^{(4)} B_b^{(4)}] &= \mathbb{E}\left[a B_{\frac{1}{a}} \cdot b B_{\frac{1}{b}}\right] \\ &= ab \cdot \mathbb{E}\left[B_{\frac{1}{a}} B_{\frac{1}{b}}\right] \end{aligned}$$

From 2.1.i).c), we have:

$$\begin{aligned} \mathbb{E}[B_a^{(4)} B_b^{(4)}] &= ab \cdot \min\left(\frac{1}{a}, \frac{1}{b}\right) \\ &= \min\left(\frac{ab}{a}, \frac{ab}{b}\right) \\ &= \min(a, b) \end{aligned}$$

Thus, 2.1.i).c) is satisfied.

Therefore, by 2.1.ii), we conclude that $B^{(4)}$ is a Brownian motion.

□

3 Itô integral

This section (and the following one) will introduce the notions of Itô calculus that we will need to model the optimal stopping time problem.

The aim of this section is to define the Itô integral and give some of its properties.

Before defining the Itô integral we need to define the class of functions on which it will be defined.

Définition 3.1. Let $V = V(S, T)$ be the class of functions

$$f(t, \omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R}$$

such that:

- (i) The mapping $(t, \omega) \mapsto f(t, \omega)$ is $\mathcal{B} \times \mathcal{F}$ -measurable, where $\mathcal{B}$ denotes the Borel σ -algebra on $[0, \infty)$.
- (ii) $f(t, \omega)$ is $\mathcal{F}_t$ -adapted.
- (iii) $\mathbb{E} \left[\int_S^T f(t, \omega)^2 dt \right] < \infty$.

For functions $f \in V$, we will now show how to define the Itô integral :

$$I[f](\omega) = \int_S^T f(t, \omega) dB_t(\omega),$$

where B_t is a one-dimensional Brownian motion.

We will procede in a natural way : First, we define $I[\varphi]$ for a simple class of functions φ . Then we show that each $f \in V$ can be approximated (in an appropriate sense) by such φ 's, and we use this to define $\int f dB$ as the limit of $\int \varphi dB$ as $\varphi \rightarrow f$.

Definition 3.2 (Elementary function). *A function $\varphi \in V$ is called **elementary** if it has the form :*

$$\varphi(t, \omega) = \sum_j e_j(\omega) \cdot \mathbf{1}_{[t_j, t_{j+1})}(t). \quad (3.1)$$

Remark 3.1. Since $\varphi \in V$, each function e_j must be $\mathcal{F}_{t_j}$ -measurable.

Définition 3.3. For elementary functions $\varphi(t, \omega)$, we define the Itô integral as such :

$$\int_S^T \varphi(t, \omega) dB_t(\omega) = \sum_{j \geq 0} e_j(\omega) (B_{t_{j+1}} - B_{t_j})(\omega) \quad (3.2)$$

Now we make the following important observation:

Lemma 3.1 (Itô Isometry). *If $\varphi(t, \omega)$ is bounded and elementary, then:*

$$\mathbb{E} \left[\left(\int_S^T \varphi(t, \omega) dB_t(\omega) \right)^2 \right] = \mathbb{E} \left[\int_S^T \varphi(t, \omega)^2 dt \right]. \quad (3.3)$$

Proof. Put $\Delta B_j = B_{t_{j+1}} - B_{t_j}$. Then:

$$\mathbb{E}[e_i e_j \Delta B_i \Delta B_j] = \begin{cases} 0, & \text{if } i \neq j, \\ \mathbb{E}[e_j^2] \cdot (t_{j+1} - t_j), & \text{if } i = j, \end{cases}$$

using the fact that $e_i e_j \Delta B_i$ and ΔB_j are independent if $i < j$. Thus:

$$\mathbb{E} \left[\left(\int_S^T \varphi dB \right)^2 \right] = \sum_{i,j} \mathbb{E}[e_i e_j \Delta B_i \Delta B_j] = \sum_j \mathbb{E}[e_j^2] \cdot (t_{j+1} - t_j) = \mathbb{E} \left[\int_S^T \varphi^2 dt \right].$$

□

The idea is now to use the isometry (3.3) to extend the definition from elementary functions to functions in $\mathcal{V}$.

We will do this in several steps :

Step 1. Let $g \in \mathcal{V}$ be bounded and $g(\cdot, \omega)$ continuous for each ω . Then there exist elementary functions $\varphi_n \in \mathcal{V}$ such that :

$$\mathbb{E} \left[\int_S^T (g - \varphi_n)^2 dt \right] \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Define:

$$\varphi_n(t, \omega) = \sum_j g(t_j, \omega) \cdot \chi_{[t_j, t_{j+1})}(t).$$

Then φ_n is elementary since $g \in \mathcal{V}$, and we have :

$$\int_S^T (g - \varphi_n)^2 dt \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

for each ω , since $g(\cdot, \omega)$ is continuous. Hence, by the bounded convergence theorem:

$$\mathbb{E} \left[\int_S^T (g - \varphi_n)^2 dt \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

□

Step 2. Let $h \in \mathcal{V}$ be bounded. Then there exist bounded functions $g_n \in \mathcal{V}$ such that $g_n(\cdot, \omega)$ is continuous for all ω and n , and :

$$\mathbb{E} \left[\int_S^T (h - g_n)^2 dt \right] \rightarrow 0.$$

Proof. Suppose $|h(t, \omega)| \leq M$ for all (t, ω) . For each n , let ψ_n be a non-negative, continuous function on $\mathbb{R}$ such that:

- (i) $\psi_n(x) = 0$ for $x \leq -\frac{1}{n}$ and $x \geq 0$,

$$(ii) \int_{-\infty}^{\infty} \psi_n(x) dx = 1.$$

Define :

$$g_n(t, \omega) = \int_0^t \psi_n(s-t) h(s, \omega) ds.$$

Then $g_n(\cdot, \omega)$ is continuous for each ω , and $|g_n(t, \omega)| \leq M$. Since $h \in \mathcal{V}$, it is possible to show that $g_n(t, \cdot)$ is $\mathcal{F}_t$ -measurable for all t .

Moreover :

$$\int_S^T (g_n(s, \omega) - h(s, \omega))^2 ds \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad \text{for each } \omega.$$

We have, by bounded convergence :

$$\mathbb{E} \left[\int_S^T (h(t, \omega) - g_n(t, \omega))^2 dt \right] \xrightarrow{n \rightarrow \infty} 0,$$

as asserted. □

Step 3. Let $f \in \mathcal{V}$. Then there exists a sequence $\{h_n\} \subset \mathcal{V}$ such that each h_n is bounded and

$$\mathbb{E} \left[\int_S^T (f - h_n)^2 dt \right] \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Define:

$$h_n(t, \omega) = \begin{cases} -n, & \text{if } f(t, \omega) < -n, \\ f(t, \omega), & \text{if } -n \leq f(t, \omega) \leq n, \\ n, & \text{if } f(t, \omega) > n. \end{cases}$$

Then the conclusion follows directly from the dominated convergence theorem. □

We now have an efficient way to approximate all elements of $\mathcal{V}$.

Thanks to that we can now give a proper definition to :

$$\int_S^T f(t, \omega) dB_t(\omega), \quad f \in \mathcal{V}.$$

If $f \in \mathcal{V}$, we choose, by Steps 1–3, elementary functions $\varphi_n \in \mathcal{V}$ such that:

$$\mathbb{E} \left[\int_S^T |f - \varphi_n|^2 dt \right] \xrightarrow{n \rightarrow \infty} 0.$$

Then define:

$$I[f](\omega) := \int_S^T f(t, \omega) dB_t(\omega) := \lim_{n \rightarrow \infty} \int_S^T \varphi_n(t, \omega) dB_t(\omega).$$

The limit exists as an element of $L^2(P)$, since the sequence $\int_S^T \varphi_n(t, \omega) dB_t(\omega)$ forms a Cauchy sequence in $L^2(P)$, by the Itô isometry 3.1.

This lead us to this definition :

Definition 3.4 (Itô integral on $\mathcal{V}$). Let $f \in \mathcal{V}(S, T)$.
Then the Itô integral of f (from S to T) is defined by:

$$\int_S^T f(t, \omega) dB_t(\omega) = \lim_{n \rightarrow \infty} \int_S^T \varphi_n(t, \omega) dB_t(\omega) \quad (\text{limit in } L^2(P)),$$

where $\{\varphi_n\}$ is a sequence of elementary functions such that:

$$\mathbb{E} \left[\int_S^T (f(t, \omega) - \varphi_n(t, \omega))^2 dt \right] \xrightarrow{n \rightarrow \infty} 0.$$

Moreover, thanks to the Itô isometry (3.1), the limit doesn't depend on the choice of $\{\varphi_n\}$.
Furthermore, from the Itô isometry and by definition of the Itô integral on $\mathcal{V}$ we get the following important result :

Corollary 3.1 (The Itô isometry (on $\mathcal{V}$)).

$$\mathbb{E} \left[\left(\int_S^T f(t, \omega) dB_t \right)^2 \right] = \mathbb{E} \left[\int_S^T f^2(t, \omega) dt \right], \quad \text{for all } f \in \mathcal{V}(S, T). \quad (3.1.14)$$

Corollary 3.2. If $f(t, \omega) \in \mathcal{V}(S, T)$ and $\{f_n(t, \omega)\} \subset \mathcal{V}(S, T)$ for $n = 1, 2, \dots$, and if:

$$\mathbb{E} \left[\int_S^T (f_n(t, \omega) - f(t, \omega))^2 dt \right] \xrightarrow{n \rightarrow \infty} 0,$$

then :

$$\int_S^T f_n(t, \omega) dB_t(\omega) \longrightarrow \int_S^T f(t, \omega) dB_t(\omega) \quad \text{in } L^2(P) \text{ as } n \rightarrow \infty.$$

The extra term $-\frac{1}{2}t$ shows that the Itô stochastic integral does not behave like ordinary integrals.

Theorem 3.1. Let $f, g \in \mathcal{V}(0, T)$ and let $0 \leq S < U < T$.

Then :

(i)

$$\int_S^T f dB_t = \int_S^U f dB_t + \int_U^T f dB_t, \quad \text{for a.a. } \omega.$$

(ii)

$$\int_S^T (cf + g) dB_t = c \cdot \int_S^T f dB_t + \int_S^T g dB_t, \quad (c \text{ constant}), \quad \text{for a.a. } \omega.$$

(iii)

$$\mathbb{E} \left[\int_S^T f dB_t \right] = 0.$$

(iv) $\int_S^T f dB_t$ is $\mathcal{F}_T$ -measurable.

Proof. This clearly holds for all elementary functions, so by taking limits we obtain this for all $f, g \in \mathcal{V}(0, T)$. $\square$

Theorem 3.2. *Let $f(t, \omega) \in \mathcal{V}(0, T)$ for all T . Then the process defined by:*

$$M_t(\omega) = \int_0^t f(s, \omega) dB_s$$

is a martingale with respect to $\{\mathcal{F}_t\}$.

We admit this result.

4 Itô formula

We saw Itô integrals in the previous section but similarly to Riemann's integrals, using the definition is not very convenient for computing the value of a given integral. Finding an easier way is the topic of this section.

Instead of trying to find a general result, we will focus on a specific set of Itô integrals. For this, we will introduce the following definition and try to compute the Itô integral appearing in the right term.

Definition 4.1 (1-dimensional Itô processes). *Let B_t be a 1-dimensional Brownian motion on $(\Omega, \mathcal{F}, P)$. A (1-dimensional) Itô process (or stochastic integral) is a stochastic process X_t on $(\Omega, \mathcal{F}, P)$ of the form :*

$$X_t = X_0 + \int_0^t u(s, \omega) ds + \int_0^t v(s, \omega) dB_s, \quad (4.1)$$

where $v \in \mathcal{V}$, so that :

$$P\left(\int_0^t v(s, \omega)^2 ds < \infty \text{ for all } t \geq 0\right) = 1,$$

We also assume that u is $\mathcal{H}_t$ -adapted (where $\mathcal{H}_t$ is the natural filtration of X_t) and :

$$P\left(\int_0^t |u(s, \omega)| ds < \infty \text{ for all } t \geq 0\right) = 1.$$

If X_t is an Itô process of the form (4.1), the equation (4.1) is sometimes written in the shorter differential form :

$$dX_t = u dt + v dB_t$$

We can now state a core result of the Itô calculus : the Itô formula. This formula will allow us to more easily calculate integrals.

Theorem 4.1 (Itô formula). *Let X_t be an Itô process given by:*

$$dX_t = u dt + v dB_t.$$

Let $g(t, x) \in C^2([0, \infty) \times \mathbb{R})$. Then:

$$Y_t = g(t, X_t)$$

is again an Itô process, and :

$$dY_t = \frac{\partial g}{\partial t}(t, X_t) dt + \frac{\partial g}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2}(t, X_t) \cdot (dX_t)^2.$$

Example 4.1. What is :

$$\int_0^t s dB_s ?$$

If we use classical (usual) calculus intuition, it seems reasonable that a term of the form tB_t should appear, so we put :

$$g(t, x) = tx$$

and define :

$$Y_t = g(t, B_t) = tB_t.$$

Then, by Itô's formula:

$$dY_t = B_t dt + t dB_t + 0,$$

i.e.

$$d(tB_t) = B_t dt + t dB_t.$$

Integrating both sides from 0 to t , we obtain:

$$tB_t = \int_0^t B_s ds + \int_0^t s dB_s.$$

5 Application

5.1 Model and notations

Now we have (almost) all the tools we need.

We will consider the following situation : we want to find the best investing strategy in the stock market. What we mean by strategy is "when to buy and when to sell". To simplify the problem we want to find the best two values : a max and a min, where we will sell our stocks.

The first challenge is to find a good model for the situation.

If we call S_t the price of our stock at the time t , we can model the situation by saying that S_t follows this equation :

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where W_t is a standard Brownian motion.

In other words, the price depends on a general trend symbolized by $\mu S_t dt$: the drift, and on random variation symbolized by $\sigma S_t dW_t$.

We call :

- μ : the drift
- σ : the volatility

By using the Itô formula, we can solve this equation.

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad X_0 > 0, \quad \mu, \sigma \in \mathbb{R}, \quad \sigma \geq 0. \quad (5.1)$$

For $g(t, x) = g(x) = \ln x$ we have $g'(x) = 1/x$ and $g''(x) = -1/x^2$. Itô's formula in one dimension gives :

$$\begin{aligned} dY_t &= d(\ln S_t) = 0 + g'(S_t) dS_t + \frac{1}{2} g''(S_t) (dS_t)^2 \\ &= \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (dS_t)^2. \end{aligned} \quad (5.2)$$

From (5.1), $dS_t = \mu S_t dt + \sigma S_t dW_t$. Using Itô rules $(dW_t)^2 = dt$, $dt dW_t = 0$, $(dt)^2 = 0$, we get

$$(dS_t)^2 = \sigma^2 S_t^2 dt. \quad (5.3)$$

Plugging dX_t and (5.3) into (5.2) :

$$\begin{aligned} d(\ln S_t) &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) - \frac{1}{2} \frac{1}{S_t^2} (\sigma^2 S_t^2 dt) \\ &= \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \end{aligned} \quad (5.4)$$

Integrate (5.4) from 0 to t :

$$\ln S_t - \ln S_0 = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t. \quad (5.5)$$

Exponentiating (5.5) yields the explicit solution

$$\boxed{S_t = S_0 \exp\left(\left(\mu - \frac{1}{2} \sigma^2\right) t + \sigma W_t\right)}. \quad (5.6)$$

Remark 5.1. This kind of Brownian motion is called a **Geometric Brownian Motion (GBM)**

Remark 5.2. Another equation worth noting is this one :

$$dX_t = \hat{\mu} dt + \hat{\sigma} X_t dW_t$$

If we solve it, we get :

$$\boxed{X_t = X_0 + \hat{\mu} t + \hat{\sigma} W_t}$$

which is called an **Arithmetic Brownian Motion (ABM)**.

So if we adjust the parameters, we have :

$$\frac{S_t}{S_0} = \exp(X_t - X_0)$$

With the link given above, we can see that getting results on one gives results on the other. We will work with Geometric Brownian Motion (GBM) and consider the following model of our situation :

Let

$$T = T_{-a} \wedge T_b$$

where $a, b \in \mathbb{R}_+$ and :

$$T_{-a} = \inf \left\{ t \geq 0, \frac{S_t}{S_0} = \exp(-a) \right\}$$

$$T_b = \inf \left\{ t \geq 0, \frac{S_t}{S_0} = \exp(b) \right\}$$

Remark 5.3. We use $\exp(-a)$ instead of a number between 0 and 1 to have boundaries in $\mathbb{R}$ and also to simplify the following calculus.

T , T_{-a} , and T_b are stopping times, representing when we sell our stocks.

We can evaluate the average performance of our boundaries a and b by calculating Φ :

$$\Phi := \mathbb{E} \left[\sum_{n=0}^{\infty} e^{-\rho T_{n+1}} \left\{ U_{\gamma} \left(\frac{S_{T_{n+1}}}{S_{T_n}} \right) - c \right\} \right],$$

where

- U_{γ} is the CRRA (Constant Relative Risk Aversion) utility function : $U_{\gamma}(x) = \frac{1}{1-\gamma} x^{1-\gamma}$
- $\gamma > 0$ is called the risk-aversion parameter,
- $\rho > 0$: discounting rate parameter (inflation),
- $c > 0$: transaction cost parameter,
- $(T_n)_{n=0}^{\infty}$: increasing sequence of stopping times such that $0 = T_0 \leq T_1 = T \leq \dots \leq T_n \leq T_{n+1} \leq \dots$. In particular, for given $a, b > 0$, we let $T_0 = 0$ and define

$$T_{n+1} := \inf \left\{ t \geq T_n \mid S_t = e^{-a} S_{T_n} \text{ or } S_t = e^b S_{T_n} \right\}, \quad (n = 0, 1, \dots).$$

We deduce the relation:

$$\Phi = \frac{1}{1 - \mathbb{E}[e^{-\rho T_1}]} \left\{ \mathbb{E} \left[e^{-\rho T_1} U_{\gamma} \left(\frac{S_{T_1}}{S_0} \right) \right] - c \mathbb{E}[e^{-\rho T_1}] \right\}$$

Indeed, using the strong Markov property and the independent increment property of Brownian motion, we see that

$$\begin{aligned} \Phi &= \mathbb{E} \left[e^{-\rho T_1} \left\{ U_{\gamma} \left(\frac{S_{T_1}}{S_0} \right) - c \right\} \right] + \mathbb{E} \left[e^{-\rho T_1} \mathbb{E} \left[\sum_{n=1}^{\infty} e^{-\rho(T_{n+1}-T_1)} \left\{ U_{\gamma} \left(\frac{S_{T_{n+1}}}{S_{T_n}} \right) - c \right\} \middle| \mathcal{F}_{T_1} \right] \right] \\ &= \mathbb{E} \left[e^{-\rho T_1} U_{\gamma} \left(\frac{S_{T_1}}{S_0} \right) \right] + \mathbb{E} [e^{-\rho T_1}] (\Phi - c). \end{aligned}$$

Define :

$$L(\rho, \gamma) = E \left[e^{-\rho T - \gamma X_T} \right]$$

where

$$X_T = \ln \frac{S_T}{S_0} = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t$$

This way we have :

$$U_\gamma \left(\frac{S_{T_1}}{S_0} \right) = \frac{1}{1 - \gamma} e^{(1 - \gamma) X_{T_1}},$$

so now we can express Φ in this new form :

$$\Phi = \Phi(a, b) = \frac{1}{1 - L(\rho, 0)} \left\{ \frac{L(\rho, \gamma - 1)}{1 - \gamma} - c L(\rho, 0) \right\}$$

Our next objective is to find an explicit value for L .

Remark 5.4. X_t is an ABM.

We have :

$$\begin{aligned} T_{-a} &= \inf \{ t \geq 0, X_t = -a \} \\ T_b &= \inf \{ t \geq 0, X_t = b \} \end{aligned}$$

Using the ABM point of view here is easier. We will show results on easy ABM then generalize for every ABM then switch back to GBM.

Remark 5.5. L is the joint Laplace transform of the time and place of stopping of X_t .

Remark 5.6. (i) If $\gamma > 1$, then, $\Phi < 0$.

(ii) If $0 < \gamma < 1$, then, $L(\rho, \gamma - 1) > L(\rho, 0)$. So, if the cost $c > 0$ is sufficiently small, (eg., if $c(1 - \gamma) < 1$ holds), then, $\Phi > 0$.

5.2 Expression of L

5.2.1 Case of a Brownian Motion

Proposition 5.1. Let X be a Brownian motion starting from 0 i.e $\mu - \frac{\sigma^2}{2} = 0$ and let

$$T = T_{-a} \wedge T_b.$$

with :

$$\begin{aligned} T_{-a} &= \inf \{ t \geq 0, X_t = -a \} \\ T_b &= \inf \{ t \geq 0, X_t = b \} \end{aligned}$$

Then the Laplace transform of the time of stop T is :

$$\mathbb{E}_0[\exp(-\rho T)] = \frac{\cosh(\sqrt{2\rho} \frac{a+b}{2})}{\cosh(\sqrt{2\rho} \frac{b-a}{2})}.$$

Proof. Applying Doob's optional sampling theorem to the bounded martingale :

$$\exp\left(\sqrt{2\rho}\left(X_{t\wedge T} - \frac{b-a}{2}\right) - \rho(t \wedge T)\right),$$

and using the fact that T is almost surely finite, we get :

$$\exp\left(-\sqrt{2\rho}\frac{b-a}{2}\right) = \mathbb{E}\left[\exp\left(\sqrt{2\rho}\left(X_T - \frac{b-a}{2}\right) - \rho T\right)\right].$$

Now, split the expectation on the disjoint events $\{T = T_b\}$ and $\{T = T_{-a}\}$:

$$\exp\left(-\sqrt{2\rho}\frac{b-a}{2}\right) = \exp\left(\sqrt{2\rho}\frac{b+a}{2}\right) \mathbb{E}[e^{-\rho T} \mathbf{1}_{\{T=T_b\}}] + \exp\left(\sqrt{2\rho}\frac{-a-b}{2}\right) \mathbb{E}[e^{-\rho T} \mathbf{1}_{\{T=T_{-a}\}}].$$

Using the symmetry of the Brownian motion ($X \mapsto -X$), a similar equation yields a linear system for the two unknowns :

$$A = \mathbb{E}[e^{-\rho T} \mathbf{1}_{\{T=T_b\}}], \quad B = \mathbb{E}[e^{-\rho T} \mathbf{1}_{\{T=T_{-a}\}}].$$

Solving gives :

$$A = \frac{\sinh(\sqrt{2\rho}a)}{\sinh(\sqrt{2\rho}(b+a))}, \quad B = \frac{\sinh(\sqrt{2\rho}b)}{\sinh(\sqrt{2\rho}(b+a))}.$$

Finally, summing the two expectations leads to the announced result :

$$\mathbb{E}_0[e^{-\rho T}] = A + B = \frac{\cosh(\sqrt{2\rho}\frac{b-a}{2})}{\cosh(\sqrt{2\rho}\frac{b+a}{2})}.$$

□

By considering $X_t - x$, $a = a + x$ and $b = b - x$, we can generalize this result for a Brownian motion starting in x .

Remark 5.7. The proposition (5.1) gives us the value of $L(\rho, 0)$ for $\mu = \frac{\sigma^2}{2}$.

Using this result, we will generalize for a BM with drift starting in x :

For $x, \lambda \in \mathbb{R}$, let

$$X_t^{(x,\lambda)} := x + W_t + \lambda t.$$

We write $X_t^{(\lambda)} := X_t^{(0,\lambda)}$ and $X_t := X_t^{(0,0)}$.

We will also adapt our notations in consequence :

For $a \in \mathbb{R}$, let

$$T_a^{(\lambda)} := \inf \left\{ t \geq 0 \mid X_t^{(\lambda)} = a \right\},$$

5.2.2 Case of a Brownian Motion with Drift

For $\lambda \in \mathbb{R}$ and $a, b > 0$. We deduce the following.

$$\mathbb{E} \left[e^{-\rho T_b^{(\lambda)}} 1_{\{T_b^{(\lambda)} < T_{-a}^{(\lambda)}\}} \right] = \mathbb{E} \left[\exp \left\{ \lambda W_{T_b^{(0)}} - \left(\rho + \frac{\lambda^2}{2} \right) T_b^{(0)} \right\} 1_{\{T_b^{(0)} < T_{-a}^{(0)}\}} \right]. \quad (5.7)$$

Indeed, using the Cameron-Martin-Maruyama-Girsanov's theorem, we define the probability measure $\tilde{\mathbb{P}}_T$ on $(\Omega, \mathcal{F}_T)$ ($T > 0$, a given constant) by

$$d\tilde{\mathbb{P}}_T|_{\mathcal{F}_t} = e^{\lambda W_t - \frac{\lambda^2}{2}t} d\mathbb{P}|_{\mathcal{F}_t} \quad \text{for } t \in [0, T]$$

to see that the process $\tilde{W}_t := W_t - \lambda t$ ($t \in [0, T]$) is a $(\tilde{\mathbb{P}}_T, \mathcal{F}_t)$ -Brownian motion. Hence, we deduce the relation

$$\mathbb{E} \left[e^{-\rho (T_b^{(\lambda)} \wedge T)} 1_{\{(T_b^{(\lambda)} \wedge T) < (T_{-a}^{(\lambda)} \wedge T)\}} \right] = \mathbb{E} \left[e^{\lambda W_T - \frac{\lambda^2}{2}T} e^{-\rho (T_b^{(0)} \wedge T)} 1_{\{(T_b^{(0)} \wedge T) < (T_{-a}^{(0)} \wedge T)\}} \right]. \quad (5.8)$$

Here, the right-hand-side of the above is

$$\begin{aligned} &= \mathbb{E} \left[\mathbb{E} \left[e^{\lambda W_T - \frac{\lambda^2}{2}T} e^{-\rho (T_b^{(0)} \wedge T)} 1_{\{(T_b^{(0)} \wedge T) < (T_{-a}^{(0)} \wedge T)\}} \middle| \mathcal{F}_{T_b^{(0)} \wedge T_{-a}^{(0)} \wedge T} \right] \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[e^{\lambda W_T - \frac{\lambda^2}{2}T} \middle| \mathcal{F}_{T_b^{(0)} \wedge T_{-a}^{(0)} \wedge T} \right] e^{-\rho (T_b^{(0)} \wedge T)} 1_{\{(T_b^{(0)} \wedge T) < (T_{-a}^{(0)} \wedge T)\}} \right] \\ &= \mathbb{E} \left[\exp \left\{ \lambda W_{T_b^{(0)} \wedge T} - \left(\rho + \frac{\lambda^2}{2} \right) (T_b^{(0)} \wedge T) \right\} 1_{\{(T_b^{(0)} \wedge T) < (T_{-a}^{(0)} \wedge T)\}} \right], \end{aligned} \quad (5.9)$$

where we use the tower-property of conditional expectation and the optional sampling theorem. Combining (5.8) and (5.9), letting $T \rightarrow \infty$, and using the bounded convergence theorem, we obtain (5.7). From (5.7) and proposition (5.1), we have :

$$\mathbb{E} \left[e^{-\rho T_b^{(\lambda)}} 1_{\{T_b^{(\lambda)} < T_{-a}^{(\lambda)}\}} \right] = e^{\lambda b} \mathbb{E} \left[e^{-\left(\rho + \frac{\lambda^2}{2}\right) T_b^{(0)}} 1_{\{T_b^{(0)} < T_{-a}^{(0)}\}} \right] = \boxed{e^{\lambda b} \frac{\sinh \left(a \sqrt{2\rho + \lambda^2} \right)}{\sinh \left((a+b) \sqrt{2\rho + \lambda^2} \right)}}.$$

Similarly, we have :

$$\mathbb{E} \left[e^{-\rho T_{-a}^{(\lambda)}} 1_{\{T_{-a}^{(\lambda)} < T_b^{(\lambda)}\}} \right] = e^{-\lambda a} \frac{\sinh \left(b \sqrt{2\rho + \lambda^2} \right)}{\sinh \left((a+b) \sqrt{2\rho + \lambda^2} \right)}.$$

5.2.3 Case of a Time-Changed Brownian Motion with Drift

Let $\lambda \in \mathbb{R}$, $\sigma > 0$ and define

$$Y_t^{(y)} = y + \sigma W_t + \lambda t, \quad (t \geq 0).$$

We write $Y_t := Y_t^{(0)}$. From the scaling property of Brownian motion, we deduce

$$(Y_t)_{t \geq 0} \stackrel{\text{law}}{=} \left(X_{\sigma^2 t}^{(\lambda/\sigma^2)} \right)_{t \geq 0}.$$

Further, letting

$$\hat{T}_a := \inf \{t \geq 0 \mid Y_t = a\},$$

we deduce

$$\begin{aligned} T_a^{(\lambda/\sigma^2)} &:= \inf \left\{ t \geq 0 \mid X_t^{(\lambda/\sigma^2)} = a \right\} \\ &= \sigma^2 \inf \left\{ t \geq 0 \mid X_{\sigma^2 t}^{(\lambda/\sigma^2)} = a \right\} \\ &\stackrel{\text{law}}{=} \sigma^2 \inf \{t \geq 0 \mid Y_t = a\} =: \sigma^2 \hat{T}_a. \end{aligned}$$

Therefore, for $a, b > 0$, we observe :

$$\begin{aligned} \mathbb{E} \left[\exp \left(-\rho \hat{T}_b \right) 1_{\{\hat{T}_b < \hat{T}_{-a}\}} \right] &= \mathbb{E} \left[\exp \left(-\frac{\rho}{\sigma^2} \sigma^2 \hat{T}_b \right) 1_{\{\sigma^2 \hat{T}_b < \sigma^2 \hat{T}_{-a}\}} \right] \\ &= \mathbb{E} \left[\exp \left(-\frac{\rho}{\sigma^2} T_b^{(\lambda/\sigma^2)} \right) 1_{\{T_b^{(\lambda/\sigma^2)} < T_{-a}^{(\lambda/\sigma^2)}\}} \right] \\ &= \exp \left(\frac{\lambda}{\sigma^2} b \right) \frac{\sinh \left(\frac{a}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right)}{\sinh \left(\frac{a+b}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right)}. \end{aligned} \quad (5.10)$$

Similarly, we have :

$$\mathbb{E} \left[\exp \left(-\rho \hat{T}_{-a} \right) 1_{\{\hat{T}_{-a} < \hat{T}_b\}} \right] = \exp \left(-\frac{\lambda}{\sigma^2} a \right) \frac{\sinh \left(\frac{b}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right)}{\sinh \left(\frac{a+b}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right)}. \quad (5.11)$$

Using (5.10) and (5.11), we can compute the explicit form of

$$L(\rho, \gamma) := \mathbb{E} \left[\exp \left\{ -\rho \left(\hat{T}_{-a} \wedge \hat{T}_b \right) - \gamma Y_{\hat{T}_{-a} \wedge \hat{T}_b} \right\} \right].$$

Indeed, we have :

$$\begin{aligned} L(\rho, \gamma) &= e^{-\gamma b} \mathbb{E} \left[e^{-\rho \hat{T}_b} 1_{\{\hat{T}_b < \hat{T}_{-a}\}} \right] + e^{\gamma a} \mathbb{E} \left[e^{-\rho \hat{T}_{-a}} 1_{\{\hat{T}_{-a} < \hat{T}_b\}} \right] \\ &= \frac{1}{\sinh \left(\frac{a+b}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right)} \left[\exp \left\{ \left(\frac{\lambda}{\sigma^2} - \gamma \right) b \right\} \sinh \left(\frac{a}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right) \right. \\ &\quad \left. + \exp \left\{ - \left(\frac{\lambda}{\sigma^2} - \gamma \right) a \right\} \sinh \left(\frac{b}{\sigma^2} \sqrt{2\rho\sigma^2 + \lambda^2} \right) \right], \end{aligned}$$

Remark 5.8. This expression is very complicated, and getting results through the second derivative of L seems high impossible.

5.3 Asymptotic results

5.3.1 Asymptotic behaviour of $L(\rho, \gamma - 1)$ as $a \rightarrow \infty$

Recall

$$\sinh(x) \sim \frac{1}{2}e^x, \quad \cosh(x) \sim \frac{1}{2}e^x \quad \text{as } x \rightarrow \infty.$$

Hence, as $a \rightarrow \infty$

$$L(\rho, \gamma - 1) \sim \exp \left[\left\{ \frac{\lambda}{\sigma^2} - \delta + (1 - \gamma) \right\} b \right] + 2e^{-b\delta} \sinh(b\delta) \exp \left[- \left\{ \frac{\lambda}{\sigma^2} + \delta + (1 - \gamma) \right\} a \right],$$

where we denote

$$\delta := \frac{\sqrt{2\rho\sigma^2 + \lambda^2}}{\sigma^2} \quad \left(\geq \frac{|\lambda|}{\sigma^2} \right).$$

We consider the following four cases: (I), (II), and (III) are called less risk-averse (LRA) cases, while (IV) is called highly risk-averse (HRA) case.

(I) $0 < \gamma < 1$ case :

$$\lim_{a \rightarrow \infty} L(\rho, \gamma - 1; a, b) = \exp \left[\left\{ \frac{\lambda}{\sigma^2} - \delta + (1 - \gamma) \right\} b \right].$$

(II) $1 < \gamma < 1 + \frac{\lambda}{\sigma^2} + \delta$ case :

$$\lim_{a \rightarrow \infty} L(\rho, \gamma - 1; a, b) = \exp \left[\left\{ \frac{\lambda}{\sigma^2} - \delta + (1 - \gamma) \right\} b \right].$$

(III) $\gamma = 1 + \frac{\lambda}{\sigma^2} + \delta$ case :

$$\lim_{a \rightarrow \infty} L(\rho, \gamma - 1; a, b) = \exp \left[\left\{ \frac{\lambda}{\sigma^2} - \delta + (1 - \gamma) \right\} b \right] + 2e^{-b\delta} \sinh(b\delta).$$

(IV) $\gamma > 1 + \frac{\lambda}{\sigma^2} + \delta$ case :

$$\lim_{a \rightarrow \infty} L(\rho, \gamma - 1; a, b) = \infty.$$

In this case, we deduce

$$\lim_{a \rightarrow \infty} \Phi(a, b) = -\infty.$$

That is, a HRA investor takes a decrease in wealth very seriously, and a finite lower barrier $e^{-a} > -\infty$ should be set in this case.

5.3.2 b maximizing $\Phi(a, b)$ when $a \rightarrow \infty$

First, we focus on cases (I), (II), and (III), letting $a \rightarrow \infty$ for simplicity. Recalling that

$$\lim_{a \rightarrow \infty} L(\rho, 0; +\infty, b) = e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b}$$

We deduce the following.

(I) It holds that

$$\Phi(+\infty, b) = \left\{ \frac{1}{1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b}} - 1 \right\} \left\{ U_\gamma(e^b) - c \right\}.$$

For obtaining a reasonable solution, we impose the parameter condition $c(1 - \gamma) > 1$: that is, the transaction cost $c > 0$ is fairly high, or the risk-averse parameter $0 < \gamma < 1$ is not so small. Then,

$$\lim_{b \downarrow 0} \Phi(+\infty, b) = -\infty.$$

If $c(1 - \gamma) < 1$: that is, the transaction cost $c > 0$ is too “cheap”, or the risk-averse parameter $0 < \gamma < 1$ is small, then, we have that $\lim_{b \downarrow 0} \Phi(+\infty, b) = +\infty$.

Value of b maximizing $\Phi(+\infty, b)$?

If $c(1 - \gamma) < 1$ we have $\Phi(+\infty, b) \xrightarrow{b \rightarrow 0^+} +\infty$ so there is nothing to do, we take $b = +\infty$

If $c(1 - \gamma) > 1$: we see that

$$\Phi(+\infty, b) \underset{b \rightarrow +\infty}{=} \frac{1}{1 - \gamma} e^{-(\delta - \frac{\lambda}{\sigma^2})b} \cdot e^{(1-\gamma)b} + o(\dots)$$

So when $1 - \gamma > \delta - \frac{\lambda}{\sigma^2}$:

$$\Phi(+\infty, b) \xrightarrow{b \rightarrow +\infty} +\infty$$

Again we take $b = +\infty$

If $1 - \gamma \leq \delta - \frac{\lambda}{\sigma^2}$:

Define :

$$\begin{aligned} f(b) &:= \frac{1}{1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b}} - 1 \\ &\underset{b \rightarrow +\infty}{=} e^{-(\delta - \frac{\lambda}{\sigma^2})b} \end{aligned}$$

We have :

$$\begin{aligned} f'(b) &= \frac{-(\delta - \frac{\lambda}{\sigma^2})e^{-(\delta - \frac{\lambda}{\sigma^2})b}}{\left(1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b}\right)^2} \\ &\underset{b \rightarrow +\infty}{=} -(\delta - \frac{\lambda}{\sigma^2})e^{-(\delta - \frac{\lambda}{\sigma^2})b} \end{aligned}$$

Define :

$$\begin{aligned} g(b) &:= \frac{1}{1 - \gamma} e^{(1-\gamma)b} - c \\ &\underset{b \rightarrow +\infty}{=} \frac{1}{1 - \gamma} e^{(1-\gamma)b} \end{aligned}$$

and

$$g'(b) = e^{(1-\gamma)b}$$

So :

$$\begin{aligned}\Phi(+\infty, b)_b &= f'g + fg' \\ &\underset{b \rightarrow +\infty}{=} -\left(\delta - \frac{\lambda}{\sigma^2}\right)e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \cdot \frac{1}{1-\gamma}e^{(1-\gamma)b} + e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \cdot e^{(1-\gamma)b} \\ &\underset{b \rightarrow +\infty}{=} e^{(1-\gamma-\delta+\frac{\lambda}{\sigma^2})b} \left[1 - \frac{\delta - \frac{\lambda}{\sigma^2}}{1-\gamma}\right] + o(\dots) < 0\end{aligned}$$

Since $\Phi(+\infty, b) \xrightarrow{b \rightarrow 0^+} -\infty$, $\phi(+\infty, b)$ has a maximum in $(0, +\infty)$

Finding the exact value of b is tricky. We need to find b such that $(fg)'(b) = 0$. It's equivalent to solving the following equation :

$$\begin{aligned}f'g + fg' &= 0 \\ \Leftrightarrow \left[\frac{-(\delta - \frac{\lambda}{\sigma^2})e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b}}{\left(1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b}\right)^2} \right] \left[\frac{1}{1-\gamma}e^{(1-\gamma)b} - c \right] + \left[\frac{1}{1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b}} - 1 \right] \left[e^{(1-\gamma)b} \right] &= 0 \\ \Leftrightarrow \left[-\left(\delta - \frac{\lambda}{\sigma^2}\right)e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \right] \left[\frac{1}{1-\gamma}e^{(1-\gamma)b} - c \right] + \left[1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \right] e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \left[e^{(1-\gamma)b} \right] &= 0 \\ \Leftrightarrow \left[-\left(\delta - \frac{\lambda}{\sigma^2}\right)e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \right] \left[e^{(1-\gamma)b} - \hat{c} \right] + (1-\gamma) \left[1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \right] e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \left[e^{(1-\gamma)b} \right] &= 0 \\ \Leftrightarrow -\left(\delta - \frac{\lambda}{\sigma^2}\right) \left[e^{(1-\gamma)b} - \hat{c} \right] + (1-\gamma) \left[1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \right] \left[e^{(1-\gamma)b} \right] &= 0\end{aligned}$$

Finally we get the equation 5.12 which characterize the maximum.

$$\boxed{1 - e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} = \frac{\delta - \frac{\lambda}{\sigma^2}}{1-\gamma} \left(1 - (1-\gamma)ce^{-(1-\gamma)b}\right)} \quad (5.12)$$

We can see that a solution clearly exists : $\frac{\delta - \frac{\lambda}{\sigma^2}}{1-\gamma} > 1$ and $(1-\gamma)c > 1$, so for $b = 0$, the left term minus the right term is > 0 and for $b = +\infty$ the left term minus the right term is < 0 . We can even say that the solution of the equation is **unique** i.e. there is only one local extremum and it's the global maximum.

Indeed if we call $A(b)$ the left term and $B(b)$ the right term, then :

$$\begin{aligned}A'(b) &= \left(\delta - \frac{\lambda}{\sigma^2}\right) e^{-\left(\delta - \frac{\lambda}{\sigma^2}\right)b} \\ B'(b) &= \left(\delta - \frac{\lambda}{\sigma^2}\right) (1-\gamma)ce^{-(1-\gamma)b}\end{aligned}$$

So :

$$\frac{B'(b)}{A'(b)} = (1 - \gamma)ce^{[(\delta - \frac{\lambda}{\sigma^2}) - (1 - \gamma)]b} > 1 \text{ for all } b > 0$$

and $A' > 0$ for all b .

So $A - B$ is strictly monotonous.

We can still solve 5.12 numerically, and doing so is fairly easy since there is only one local maximum.

(II) It holds that

$$\Phi(+\infty, b) = \left\{ \frac{1}{1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b}} - 1 \right\} \left\{ U_\gamma(e^b) - c \right\}.$$

In this case, we automatically have

$$\lim_{b \downarrow 0} \Phi(+\infty, b) = -\infty.$$

Value of b maximizing $\Phi(+\infty, b)$?

The limit in 0^+ is $-\infty$, so apart from knowing that the derivative function is positive in 0^+ , it gives us no information.

We are still in case where $1 - \gamma < 0 < \delta - \frac{\lambda}{\sigma^2}$.

So we know that :

$$\Phi(+\infty, b) \xrightarrow{b \rightarrow +\infty} 0$$

The previous asymptotic equivalents are still true and so is (5.12).

We have :

$$\begin{aligned} \Phi(+\infty, b)_b &= f'g + fg' \\ &\underset{b \rightarrow +\infty}{=} -(\delta - \frac{\lambda}{\sigma^2})e^{-(\delta - \frac{\lambda}{\sigma^2})b} \cdot \frac{1}{1 - \gamma}e^{(1 - \gamma)b} + e^{-(\delta - \frac{\lambda}{\sigma^2})b} \cdot e^{(1 - \gamma)b} + o(...) \\ &\underset{b \rightarrow +\infty}{=} e^{(1 - \gamma - \delta + \frac{\lambda}{\sigma^2})b} \left[1 - \frac{\delta - \frac{\lambda}{\sigma^2}}{1 - \gamma} \right] + o(...) \end{aligned}$$

Moreover :

$$-2\delta < 1 - \gamma - \delta + \frac{\lambda}{\sigma^2} < \frac{\lambda}{\sigma^2} - \delta < 0$$

Therefore :

$$\Phi(+\infty, b)_b \xrightarrow{b \rightarrow +\infty} 0$$

which was expected...

If we go deeper, we see that the asymptotic sign of the derivative is determined by $\left[1 - \frac{\delta - \frac{\lambda}{\sigma^2}}{1 - \gamma}\right]$.

Since $1 - \gamma < 0$, the sign is always positive. Which means that asymptotically $\Phi(+\infty, b)$ is increasing. We can conjecture that's the case on $\mathbb{R}_+^*$.

We can validate this conjecture by proving that 5.12 has no solution, which is easy : the left term of 5.12 is always positive and the right term is strictly negative.

Therefore the maximum is reached, in that case, for $b = +\infty$

(III) It holds that

$$\Phi(+\infty, b) = \frac{1}{e^{-2\delta b} - e^{-(\delta - \frac{\lambda}{\sigma^2})b}} \left[e^{-(\delta - \frac{\lambda}{\sigma^2})b} \{U_\gamma(e^b) - c\} + \left(\frac{1}{1 - \gamma} - c\right) (1 - e^{-2\delta b}) \right].$$

Value of b maximizing $\Phi(+\infty, b)$?

We have :

$$(1 - \gamma) - \delta + \frac{\lambda}{\sigma^2} = -2\delta$$

and :

$$\begin{aligned} \Phi(+\infty, b) &= \frac{1}{e^{-2\delta b} - e^{-(\delta - \frac{\lambda}{\sigma^2})b}} \left[e^{-(\delta - \frac{\lambda}{\sigma^2})b} \left(\frac{e^{(1-\gamma)b}}{1 - \gamma} - c \right) + \left(\frac{1}{1 - \gamma} - c \right) (1 - e^{-2\delta b}) \right] \\ &\underset{b \rightarrow +\infty}{=} -e^{(\delta - \frac{\lambda}{\sigma^2})b} \left(\frac{1}{1 - \gamma} - c \right) + o(\dots) \\ &\xrightarrow{b \rightarrow +\infty} +\infty \end{aligned}$$

So we should take $b = +\infty$

We can summarize what we showed by this proposition :

Proposition 5.1. *When $a = +\infty$*

We either have :

- *a solution b_0 in $\mathbb{R}_+^*$ to the equation 5.13, and $b \rightarrow \Phi(+\infty, b)$ reaches its maximum in b_0 .*
- *no solution to 5.13, and the maximum to $b \rightarrow \Phi(+\infty, b)$ is $+\infty$.*

$$1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b} = \frac{\delta - \frac{\lambda}{\sigma^2}}{1 - \gamma} \left(1 - (1 - \gamma)ce^{-(1-\gamma)b} \right) \quad (5.13)$$

5.4 Results for $a < \infty$

We have tackled the case where $a = +\infty$, what about when a is finite ?

5.4.1 Some graphics

We will first draw $\Phi(a, b)$ for every cases to develop an intuition.

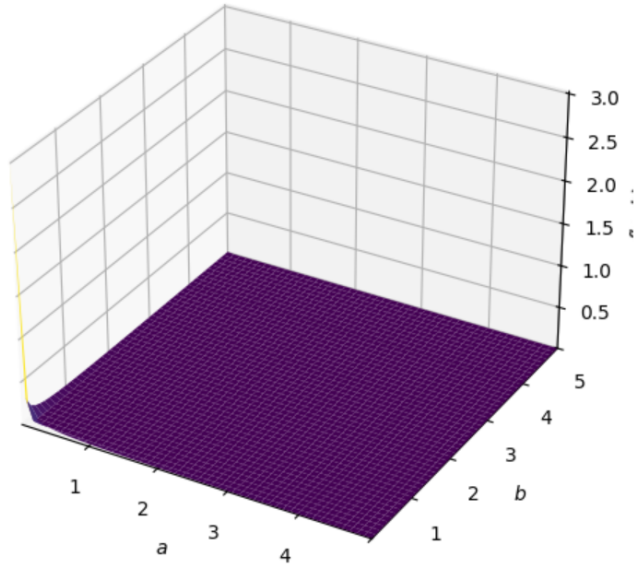


Figure 1: 1A when $c(1 - \gamma) < 1$

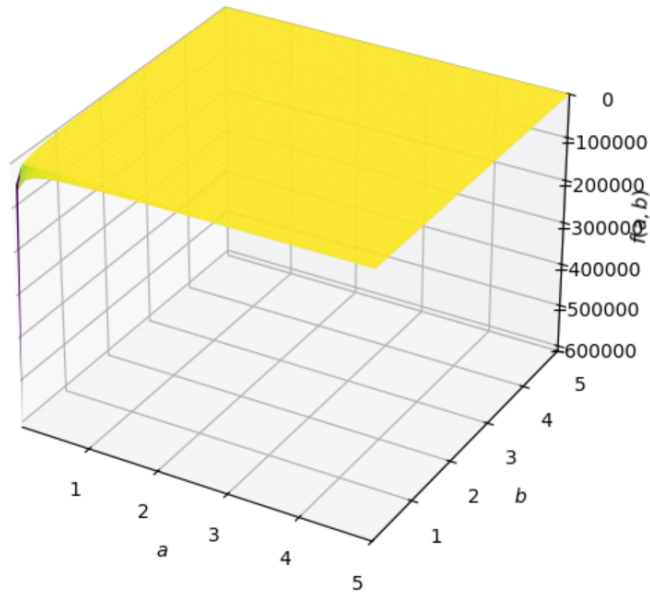


Figure 2: 1A when $c(1 - \gamma) > 1$

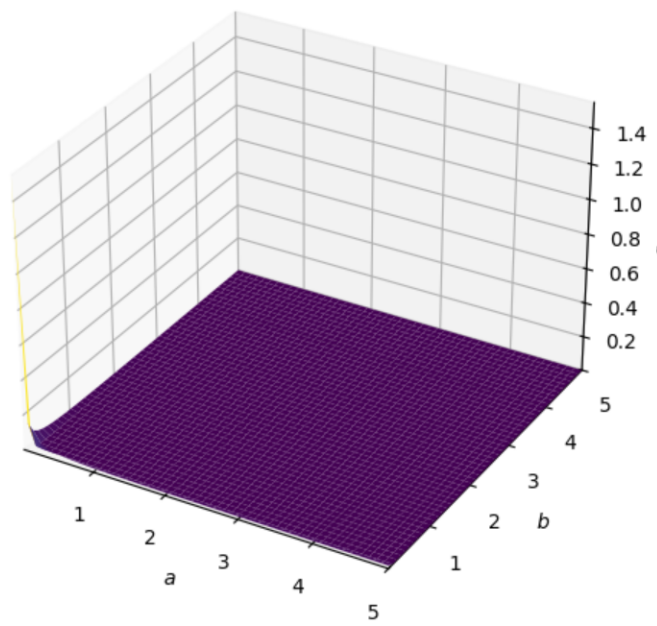


Figure 3: 1B

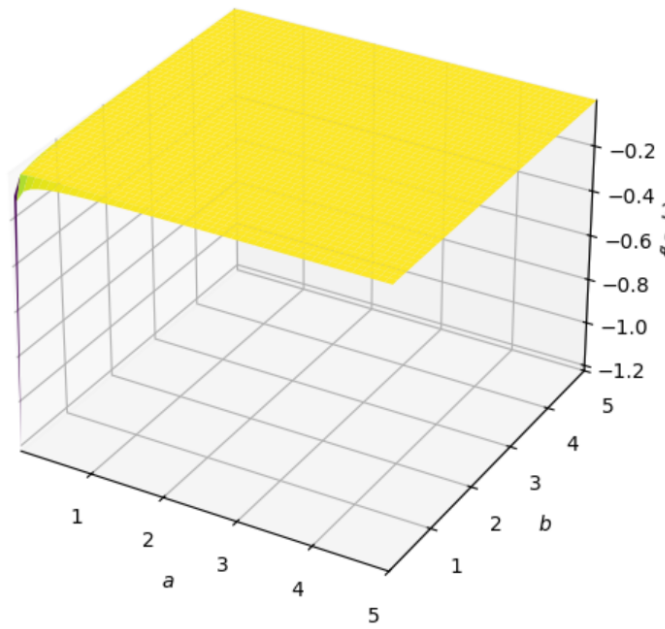


Figure 4: 2A

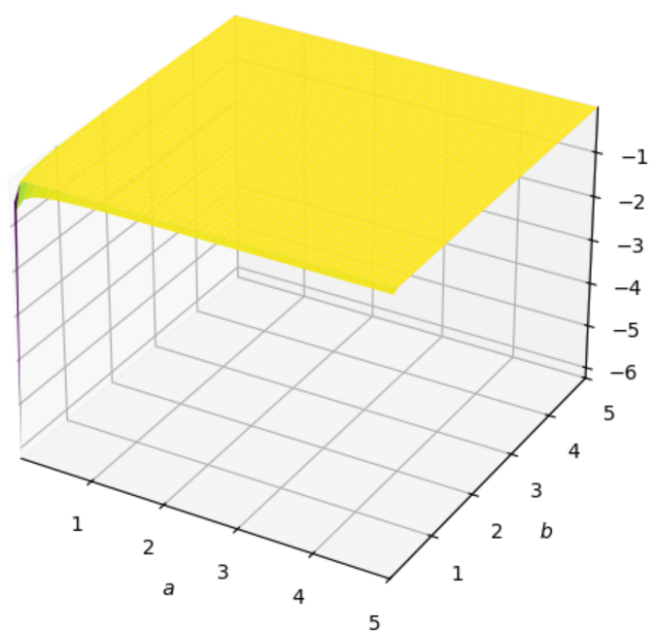


Figure 5: 2B

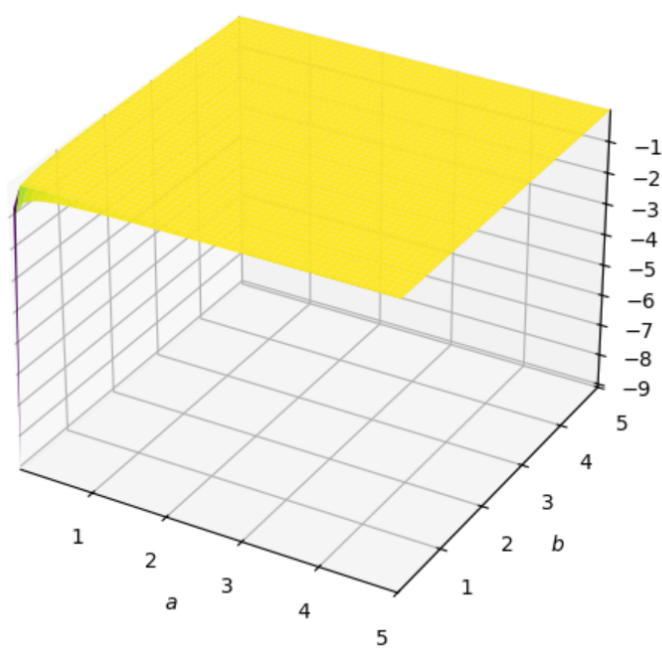


Figure 6: 3A

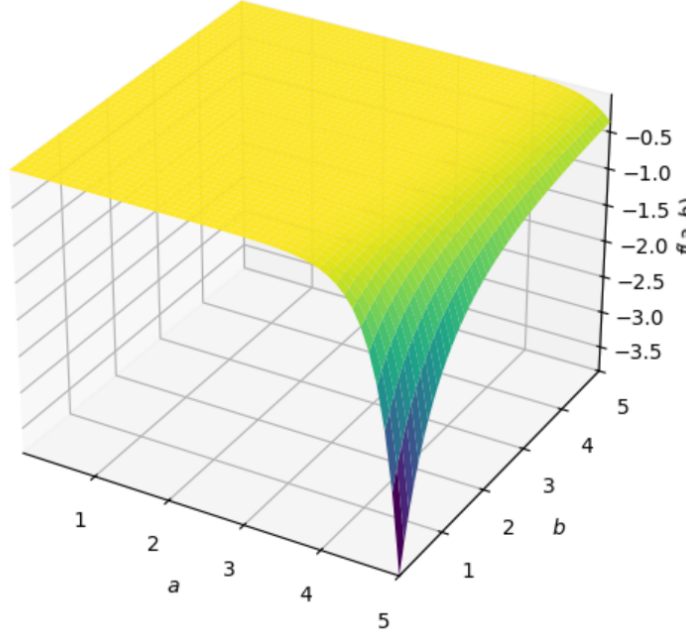


Figure 7: 3B

5.4.2 Mathematical approach

Even though I wasn't always able to find the best a and b , I tried to show, as much as possible, in this subsection the reasoning that I had had.

To lighten the expressions we may sometimes uses the following notations :

- $\hat{\gamma} = -(1 - \gamma) > -1$
- $\hat{c} = c \cdot \hat{\gamma} = c(1 - \gamma)$
- $D = \frac{1}{1-\gamma}$
- $\lambda = \mu - \frac{\sigma^2}{2}$
- $\delta = \frac{\sqrt{2\rho\sigma^2 + \lambda^2}}{\sigma^2}$

Case 1 - A) : $\lambda < 0$ and $0 < \gamma < 1$

The figures 1 and 2 shows that we have at least two subcases.

We conjecture that in $(0, 0)$ Φ is diverging.

$$\Phi = \frac{1}{1 - L(0)} \cdot \left\{ \frac{L(\gamma - 1)}{1 - \gamma} - cL(0) \right\}$$

We have $L(0) \xrightarrow{(a,b) \rightarrow (0,0)} 1^-$ since $L(0) = \mathbb{E}[e^{-\rho T}]$ and $L(\gamma) = \mathbb{E}[e^{-\rho T - \gamma X_T}] \xrightarrow{(a,b) \rightarrow (0,0)} 1$

So

$$\Phi \underset{(a,b) \rightarrow (0,0)}{=} \frac{1}{1-L(0)} \left\{ \frac{1}{1-\gamma} - c \right\} + o(\dots)$$

$$\underset{(a,b) \rightarrow (0,0)}{\longrightarrow} \begin{cases} +\infty & \text{if } c(1-\gamma) < 1 \\ -\infty & \text{if } c(1-\gamma) > 1 \end{cases}$$

Let's do the two subcases separately :

1. $c(1-\gamma) < 1$

This case is easy, since ϕ diverge toward $+\infty$ in $(0,0)$ and Φ has a finit limit in $+\infty$, the global maximum is obviously in $(0,0)$.

2. $c(1-\gamma) > 1$

- $1-\gamma > \delta - \frac{\lambda}{\sigma^2}$

Then 5.12 has no solution, $b_0 = +\infty$ and ϕ diverge toward $+\infty$ for $(a,b) \rightarrow (+\infty, +\infty)$

- $1-\gamma < \delta - \frac{\lambda}{\sigma^2}$

Then 5.12 has a solution, $b_0 < +\infty$

From now on we will focus on the second case. So :

Proposition 5.2. *The global maximum of Φ is reached in a point $(a_1, b_1) \neq (+\infty, +\infty)$*

We can try to find more information on this maximum. For exemple, figuring out if $a_0 = +\infty$. Since a and b have almost the same role in Φ we can conjecture that $a_1 \neq +\infty$. Let's work in that direction.

We can try our luck by looking at $a \mapsto \Phi(a, b_0)$. If we are lucky the maximum of this function is not $+\infty$.

We recall that b_0 is the solution of :

$$1 - e^{-(\delta - \frac{\lambda}{\sigma^2})b} = \frac{\delta - \frac{\lambda}{\sigma^2}}{1-\gamma} \left(1 - (1-\gamma)ce^{-(1-\gamma)b} \right)$$

We have :

$$\begin{aligned} \Phi(a) &= \frac{1}{1-L(0)} \cdot \left\{ \frac{L(\gamma-1)}{1-\gamma} - cL(0) \right\} \\ &= D \cdot \frac{\sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} (e^{-\hat{\gamma}b} - \hat{c}) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} (e^{\hat{\gamma}a} - \hat{c})}{\sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a}} \end{aligned}$$

Thus

$$\Phi_a(a) = D \cdot \frac{\left[\delta \cosh(a\delta)e^{\frac{\lambda}{\sigma^2}b} (e^{-\hat{\gamma}b} - \hat{c}) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} (e^{\hat{\gamma}a} - \hat{c}) \right] \left(\sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \right)}{\left(\sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \right)^2}$$

...

It doesn't look easy ! Maybe for some values of the parameters we can compute b_0 then draw $a \mapsto \Phi(a, b_0)$ and try to see what's going on.

First we need to compute the value of b_0 . We thus draw the function f where f is such that $(E) \Leftrightarrow f(b) = 0$

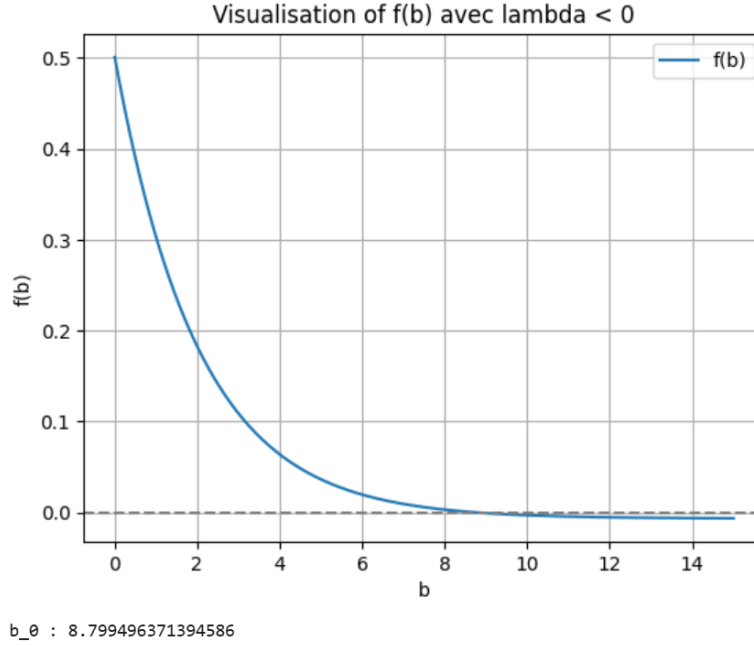


Figure 8: Graph of $f(b)$

For drawing the graphe we took :

- $\gamma = 0.5$
- $c = 3$
- $\sigma = 0.7$
- $\mu = 0.1$
- $\rho = 0.01$
- $\lambda = \mu - \frac{\sigma^2}{2} = -0.145$
- $\delta = \frac{\sqrt{2\rho\sigma^2 + \lambda^2}}{\sigma^2} \approx 0.357$
- $1 - \gamma = 0.5$
- $c(1 - \gamma) = 1.5$
- $\frac{\delta - \lambda}{\sigma^2} \approx 1.02$

I also draw it with values such that $\lambda > 0$

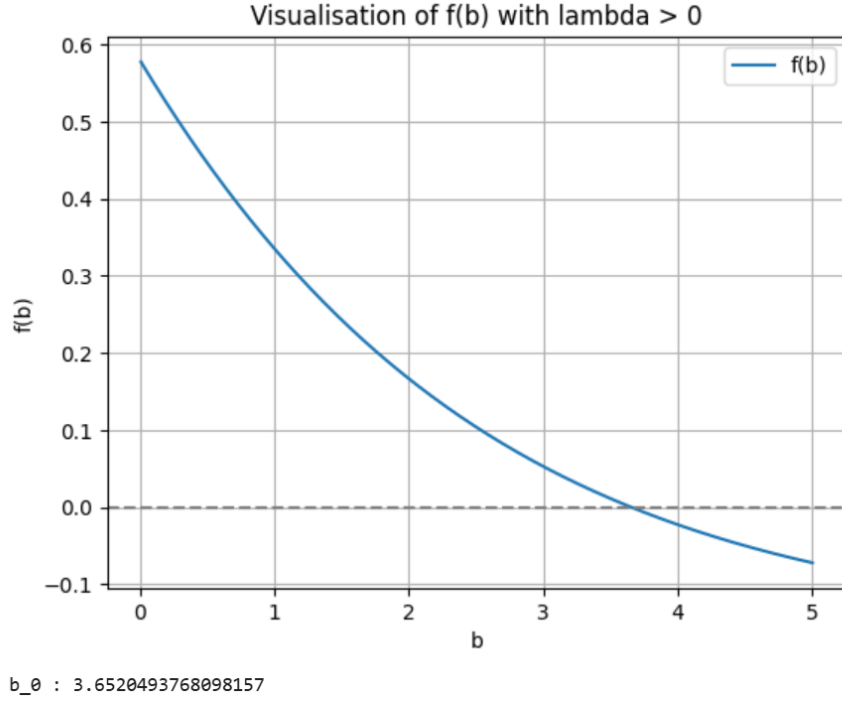


Figure 9: Graph of $f(b)$ with $\lambda > 0$

I took :

- $\gamma = 0.5$
- $c = 3$
- $\sigma = 0.5$
- $\mu = 0.3$
- $\rho = 0.01$
- $\lambda = \mu - \frac{\sigma^2}{2} = 0.175$
- $\delta = \frac{\sqrt{2\rho\sigma^2 + \lambda^2}}{\sigma^2} \approx 0.755$
- $1 - \gamma = 0.5$
- $c(1 - \gamma) = 1.5$
- $\frac{\delta - \lambda}{\sigma^2} = 2.32$

From our calculus and the graphs, the sign of λ doesn't seem to matter in the existence of a solution. However when $\lambda > 0$ it's not a maximum but a minimum.

We can now draw $a \mapsto \phi(a, b_0)$.

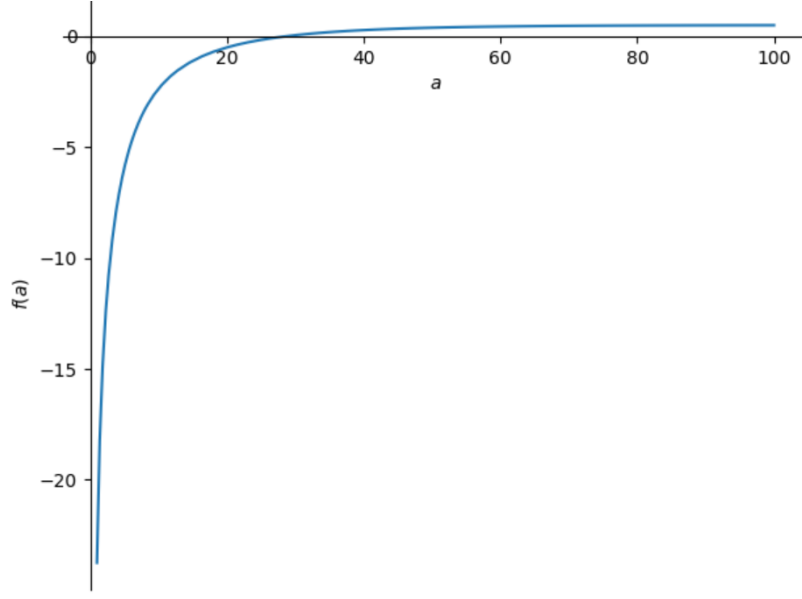


Figure 10: Graph of $a \mapsto \phi(a, b_0)$

Sadly it looks like the maximum of this function is $+\infty$ so we can't say anything on a_1 .

Conjecture 5.1. *The maximum is $(+\infty, b_0)$.*

We are back to square 1 because we still need to study the function.

Let

$$A = \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} \left(e^{-\hat{\gamma}b} - \hat{c} \right) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \left(e^{\hat{\gamma}a} - \hat{c} \right)$$

and

$$B = \sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a}$$

We have

$$\phi' = D \cdot \frac{A'B - AB'}{B^2}$$

Let

$$H = A'B - AB'$$

$$\begin{aligned} H = & \left[\delta \cosh(a\delta)e^{\frac{\lambda}{\sigma^2}b} (e^{-\hat{\gamma}b} - \hat{c}) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \left(\left(\hat{\gamma} - \frac{\lambda}{\sigma^2} \right) e^{\hat{\gamma}a} + \frac{\lambda}{\sigma^2} \hat{c} \right) \right] \\ & \times \left(\sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \right) \\ & - \left[\sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} (e^{-\hat{\gamma}b} - \hat{c}) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} (e^{\hat{\gamma}a} - \hat{c}) \right] \\ & \times \left[\delta \cosh((a+b)\delta) - \delta \cosh(a\delta)e^{\frac{\lambda}{\sigma^2}b} + \frac{\lambda}{\sigma^2} \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} \right] \end{aligned}$$

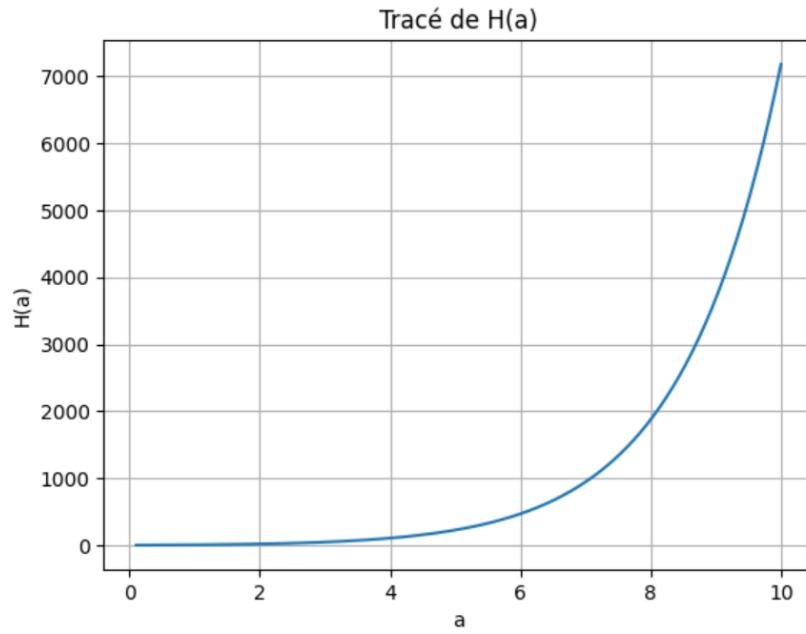


Figure 11: Graph of $H(a)$ using the previous parameters ($\lambda < 0$)

From the graphic we see that H seems convexe, sadly once again it's difficult to prove. We will thus leave this conjecture open.

Case 1 - B) : $\lambda > 0$ and $0 < \gamma < 1$

If we look at the graph of all the subcases, we don't see any difference between $\lambda < 0$ and $\lambda > 0$.

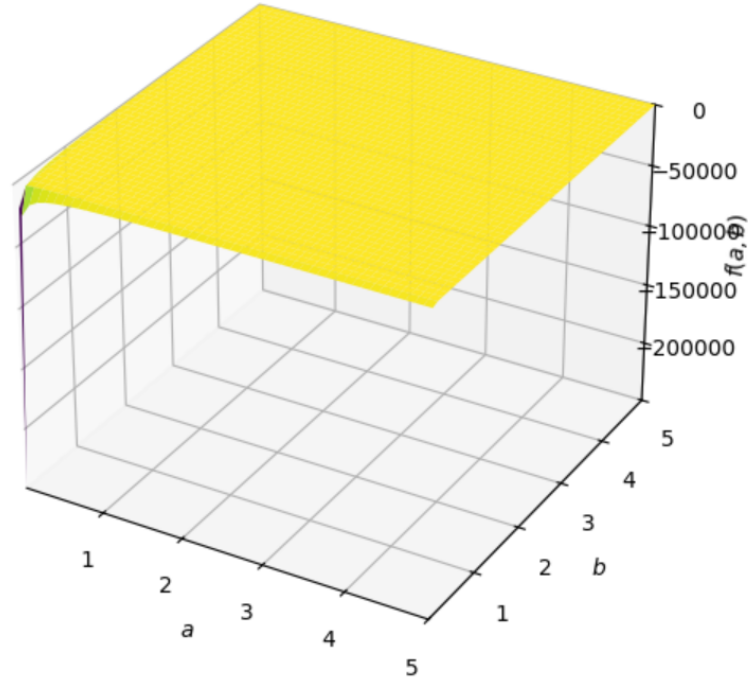


Figure 12: $\lambda > 0$, $1 - \gamma < \delta - \frac{\lambda}{\sigma^2}$, $c(1 - \gamma) > 1$

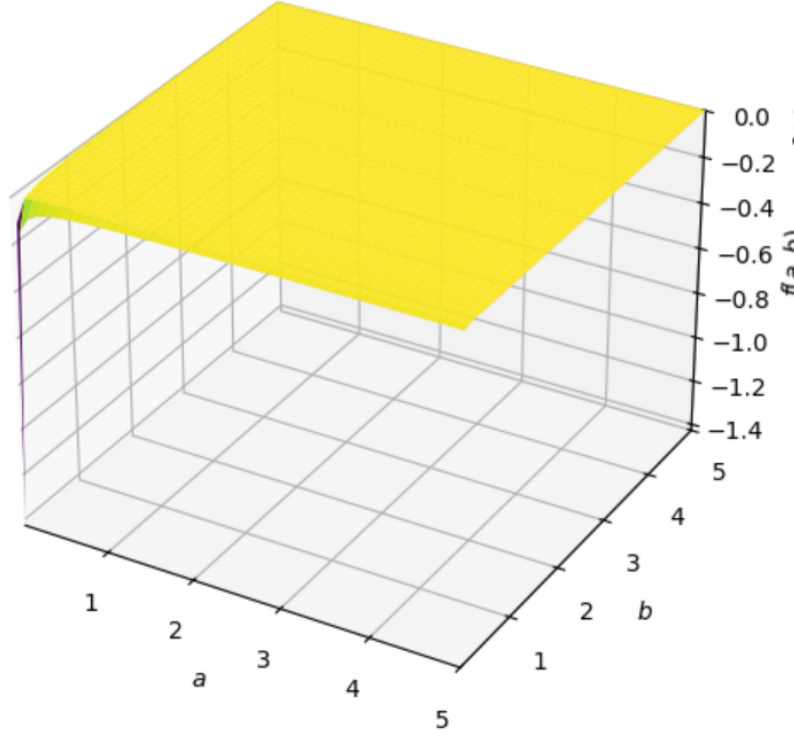


Figure 13: $\lambda < 0$, $1 - \gamma > \delta - \frac{\lambda}{\sigma^2}$, $c(1 - \gamma) < 1$

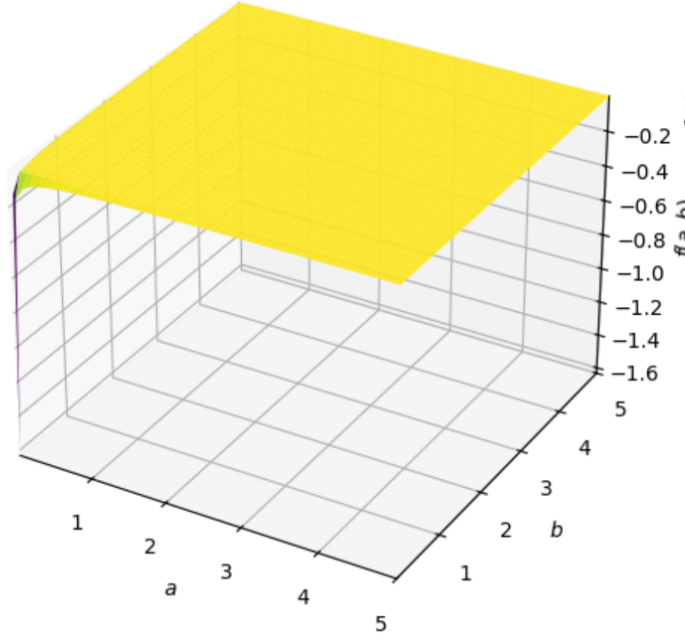


Figure 14: $\lambda > 0$, $1 - \gamma > \delta - \frac{\lambda}{\sigma^2}$, $c(1 - \gamma) < 1$

The reasoning is the same as before.

Case 2 : $1 < \gamma < 1 + \delta + \frac{\lambda}{\sigma^2}$

For now we will take $\lambda \in \mathbb{R}$,

From the graphs and what we did before, it seems the maximum is reach in $(+\infty, +\infty)$ and its value is 0.

In 0 we have $\Phi \rightarrow -\infty$. In $+\infty$ we have $\Phi \rightarrow 0$. We have two options, the first one, show that Φ is increasing. The second one is to show that ϕ is negative.

Let's try the second one.

$$\Phi(a, b) = D \cdot \frac{A}{B}$$

where

$$\begin{cases} D &= \frac{1}{1-\gamma} < 0 \\ A &= \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b}(e^{-\hat{\gamma}b} - \hat{c}) + \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a}(e^{\hat{\gamma}a} - \hat{c}) \\ B &= \sinh((a+b)\delta) - \sinh(a\delta)e^{\frac{\lambda}{\sigma^2}b} - \sinh(b\delta)e^{-\frac{\lambda}{\sigma^2}a} > 0 \end{cases}$$

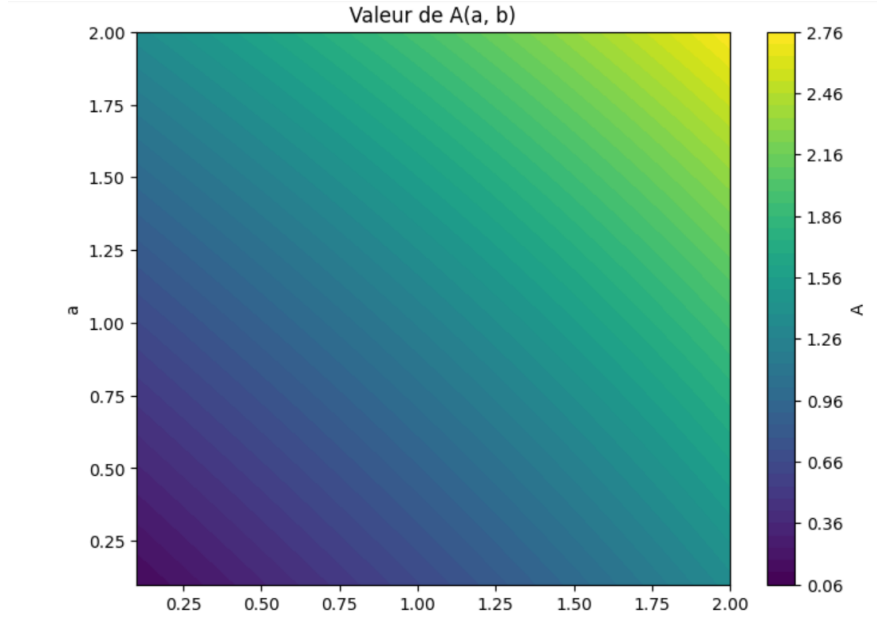


Figure 15: $A(a, b)$

We just need to show that $A > 0$

If $\hat{c} < 1$ it's obviously the case.

Sadly I don't manage to prove the case $\hat{c} \geq 1$

Case 3 : $\gamma = 1 + \delta + \frac{\lambda}{\sigma^2}$

We know from our previous work that the maximum is $+\infty$ and it's reach in $(+\infty, +\infty)$

6 Conclusion

This report was the occasion to show a part of what I have done during my internship. I was able to, on one hand learn the basis of the Itô calculus and on the other hand do 'real' research on a very precise topic.

Even though the optimal stopping time problem is not easy, this report shows that by smartly splitting the cases we can find explicit values for the best boundaries.

7 Sources

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- [2] Stochastic Differential Equations by Bernt Oksendal
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